Poisson Distribution Normal Distribution

Binomial

"Out of a fixed number of trials, how many successes do we get"

Example: Out of 100 cx, how many purchase?

Poisson

"Within a fixed interval, how many times does an event occurs?"

Example: How many cx call our support center in one hour?

Poisson Distribution

Posisson Distribution tells us the probability of a specific number of events occuring within a fixed interval of time, space, distance, area, etc, when the average rate of occurrence is known.

Technically:

Poisson Distribution is a discrete probability distribution that models the number of events occuring in a fixed interval when events occur indendently at a constant average rate.

Key Parameters:

One main parameter:

λ --> Lambda

It represents the average number of events per interval.

For eg:

A call center receives an average of 6 calls per hour.

$\lambda = 6$

The interval could be:

1 hour
1 minute
1 Day
1 km
1 square meter
1 production batch

Poisson Conditions:

1. Events occur independently

One event does not change/cause another to occur.

2. Events occur at a constant average rate.

If the average is 5 calls/hour, the model assumes this rate is reasonably stable for the interval being studied.

3. Events are counted.

The outcome is :

0,1,2,3,4,5,.....

Therefore, Poisson is discrete.

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4. Events are counted within a fixed interval

Examples

Calls per hour

Orders per minute

Defects per batch

Accidents per month

Poisson Probability formula

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

X = Poisson random variable

x = Number of events we want.

λ = Average number of events

e = Euler's Number ≈ 2.718

x! = Factorial of x 10! = 10*9*8*7*6*5*4*3*2*1

Example 1: Customer Support

A support center receives an average of 4 calls per hour.

Question

What is the probability that exactly 2 calls arrive in one hour?

Here;

λ = 4

x = 2

e = 2.718

Therefore:

P(X= 2) = (e^(-4) * 4^2)/2! 2! = 2*1 =2

0.1466

or 14.65%

0.146525

Business Interpretation

There is approximately a 14.65% probability of receiving exactly two calls in an hour when the average arrival rate is four calls per hour.

Poisson Distribution Graph

X --> Number of events

Y--> Probability

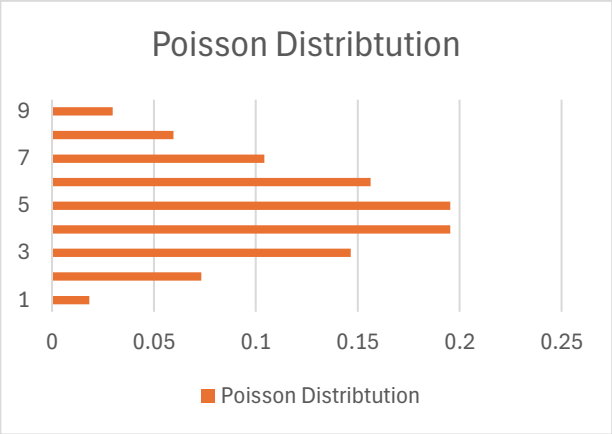
with λ = 4

0,1,2,3,4,5,6,7,8.....

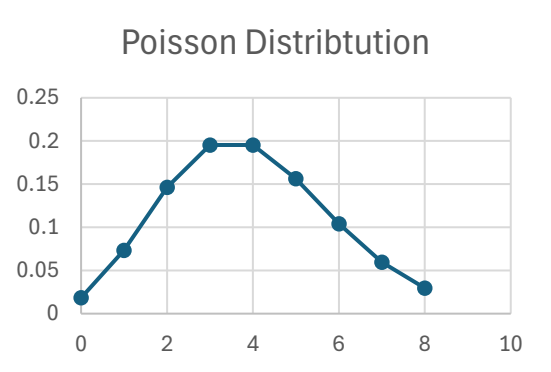
Poisson Distribution Normal Distribution

λ = 4

Evnts	Poisson Distribtution
0	0.018316
1	0.073263
2	0.146525
3	0.195367
4	0.195367
5	0.156293
6	0.104196
7	0.05954
8	0.02977



Poisson is not a continuous bell curve. It is a discrete distribution, so we represent individual evnt counts using separate probability bars.



As λ becomes larger, the distribution generally becomes more symetric and it starts looking more like a normal distribution.

Excel formula for Poisson distribution
[= POISSON.DIST(x,mean,cumulative)
cumulative --> True, False
TRUE-->2 or fewer call (at most 2 calls)
FALSE --> Exact result

Mean, Variance & SD of Poisson

For Poisson Distribution

Mean = λ
Variance = λ

Therefore:

SD = λ^(1/2)

If λ = 9
Then Mean = 9
Variance= 9
SD = 9^(1/2) 3

Mean = Variance = λ

It is the basic theoretical property of poisson distribution.

Poisson Distribution Normal Distribution

Poisson Business Usecases:

1. Customer Support

Number of calls received per hour

Useful for

- Staffing
- Workforce planning
- Peak- hour analysis

2. E-commerce

Number of orders received per minute

Useful for:

- Server capacity
- Operations planning
- Order processing

3. Manufacturing

Number of defects per production batch

Useful for

- Quality control
- Process monitoring
- Defect Analysis.

4. Logistics

Number of delivery received in an hour,.

Useful for

- Driver allocation
- Route planning
- Capcity planning (Inventory management)

5. Healthcare

Number of patients arriving at an emergency department during specific period.

Useful for:

- Staff planning
- Resource allocation
- Capacity management

Binomial		**Poisson**	
Fixed number of trials		Fixed interval	
Two outcomes per trial		Count of events	
Uses n and p		Uses λ	
Ex: Purchase among 100 cx		Ex: calls per hour	
Mean : np		Mean = λ	
Variance = np(1-p)		Variance = λ	

Binomial : fixed number of opportunities

Poisson : fixed interval of events.

Poisson Distribution Normal Distribution

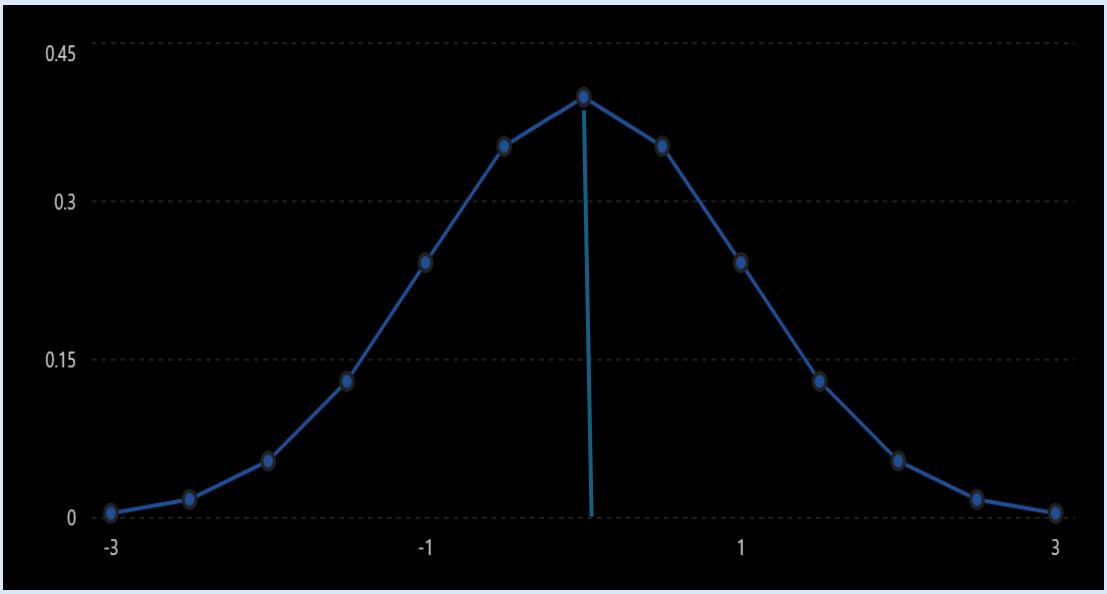
Normal Distribution

Binomial and Poisson are our discrete distributions of probability

Normal Distribution is continuous probability distribution with a bell-shaped and symmetric curve.

Technically:

A continuous probability distribution characterized by a symmetric bell-shaped curve determined by its mean and standard deviation



1. Center = Mean
2. Left and right sides are mirror images
3. **MEAN = MEADIAN = MODE**
4. Total area under the curve = 1
5. It is defined by mean and standard deviation.
6. Tails extend theoretically towards ±∞

$$Z = (X- \mu)/\sigma$$